Application of the British Athletics Muscle Injury Classification in Collegiate Football Athletes: A Retrospective, Observational Study

William Hollabaugh, MD,[†‡*] Tyler Hill, ATC,[†] Claudia Davidson, MPH,[§||] Jacquelyn Pennings, PhD,[§||¶] Nicholas Strasser, MD,[†] Lauren Porras, MD,[†] Charles Cox, MD, MPH,[†] and Robert Fitch, MD[†]

Background: Lower extremity muscle tears are common sports injuries. The British Athletics Muscle Injury Classification (BAMIC) may provide clinical guidance for tears although its value in American football and college athletes is unknown.

Hypothesis: Clinical outcomes, specifically time to return to play (RTP) (TRTP), in college American football athletes with activity-related hamstring (HS) and quadriceps (QD) tears will be associated with BAMIC.

Study Design: Retrospective cohort study.

Level of Evidence: Level 3.

Methods: American football college athletes who sustained a HS or QD tear and underwent magnetic resonance imaging within 7 days at 1 institution during the 2023 season were included. TRTP and reinjury rate (RIR) for HS and QD tears classified by BAMIC and injury characteristics were evaluated.

Results: Of the 24 HS and 10 QD tears in 21 (17.3%) of 121 athletes (20 ± 1.2 years; 100.0% male), most (60.0%) were BAMIC 2a-3b. Most (89.3%) injuries occurred before the season, and all occurred in practice. Overall median TRTP was 26 days (interquartile range, 17.8-33.0), translating to about 4 weeks missed play. The RIR was 19.0% (4/21). Most (75.0%) reinjuries occurred before RTP. There was no significant relationship between BAMIC and TRTP, although TRTP was less for grade 0 injuries and greater TRTP for injury site "c," albeit with small effect sizes.

Conclusion: Activity-related HS and QD tears are common in American football college athletes, particularly in preseason and practice. Injured athletes missed about 3 to 4 games, regardless of BAMIC, with 1 in 5 athletes suffering a reinjury. There was no relationship between BAMIC or muscle group and TRTP.

Clinical Relevance: This appraisal of BAMIC in college and American football athletes highlights the prevalence of muscle tears and the need for prevention. Although BAMIC lacked value in this study, larger studies are needed to evaluate BAMIC in this population.

Keywords: American football; British Athletics Muscle Injury Classification (BAMIC); college athlete; muscle injury/tear/strain; reinjury rate; return to play

From [†]Department of Orthopaedic Surgery, Division of Sports Medicine, Vanderbilt University Medical Center, TN, [‡]Department of Pediatrics, Vanderbilt University Medical Center, TN, [§]Department of Orthopaedic Surgery, Vanderbilt University Medical Center, TN, [||]Center for Musculoskeletal Research, Vanderbilt University Medical Center, TN, and [¶]Department of Biostatistics, Vanderbilt University School of Medicine, TN

*Address correspondence to William Louis Hollabaugh, MD, CAQSM, FAAP, Vanderbilt Sports Medicine, Vanderbilt University Medical Center, 1215 21st Avenue South, 3200 MCE South Tower, Nashville, TN 37232 (email: hollabaughw@gmail.com).

The authors report no potential conflicts of interest in the development and publication of this article.

The authors received no financial support for the research, authorship, and/or publication of this article.

DOI: 10.1177/19417381251326531

© 2025 The Author(s)

Lower extremity muscle tears are common activity-related injuries and are among the most common injuries responsible for absence from training and competition.[7,10,27] Although these injuries are often described as "strains," some clinicians prefer the term "tear" as it more accurately describes the structural characteristics of the injury.[19] Muscle tears impose a burden in sports requiring acceleration, sprinting, and kicking, including soccer, rugby, track and field, and American football (football).[1,3,7,8,15]

The muscles most injured among National Football League (NFL) athletes are the hamstring (HS), followed by the quadriceps (QD), with HS injuries ranking as the second most common injury in the NFL and the third most common injury in college football.[4,9,14] Despite their frequency, there is limited evidence defining return to play (RTP) times for HS and/or QD tears in college football athletes.[5,14,15] Also, the evidence for prognostic guidance for HS and QD injuries is constrained by the heterogeneity of injury grading and classification systems.[14]

Muscle injury classification and grading systems may be important because they may provide therapeutic and prognostic guidance. Previous studies using magnetic resonance imaging (MRI) to predict time to return to play (TRTP) in football athletes have demonstrated conflicting results, and some studies have reported that accurate prediction of TRTP is not possible using MRI classifications alone.[5,11,14,29]

In 2014, the British Athletics Muscle Injury Classification (BAMIC), an MRI-based classification designed to define muscle tears, predict clinical outcomes, and guide rehabilitation, was published (Figure 1).[23] Since its publication, some studies have shown that BAMIC may offer superior therapeutic and prognostic guidance for lower extremity muscle injuries compared with previous classification and grading systems, particularly in predicting TRTP and reinjury rate (RIR).[2,13,16-18,20-22,24-26,28,29]. Among studies assessing the clinical value of BAMIC, athletes were mostly elite or professional level and played soccer or track and field. Presently, the authors are not aware of studies that appraise the clinical value of activity-related lower extremity muscle tears classified by BAMIC in college athletes or in the context of football. Therefore, the value of BAMIC in college athletes or in the football sport is unknown, but pertinent and further investigation is warranted.

The aim of this study was to assess clinical outcomes, specifically TRTP and RIR, and injury characteristics of activity-related muscle injuries classified by BAMIC in National Collegiate Athletic Association Division-I (NCAA-DI) football athletes during the 2023 season.

METHODS

Study Design and Participants

This study was a retrospective observational cohort study with the aim of evaluating clinical outcomes, particularly TRTP and RIR, and injury characteristics of activity-related HS or QD muscle injuries as classified by BAMIC in NCAA-DI American football college athletes during the 2023 season (spring, preseason, regular season). Injuries were eligible for study inclusion if the injured athlete was an active roster football athlete during the 2023 season at the host institution, experienced an acute-onset football-related lower extremity muscle injury involving the HS and/or QD muscle group(s), received a subsequent diagnosis of a muscle tear by a team physician, underwent an MRI scan without contrast of the affected muscle group within 7 days of the date of index injury, and had the MRI scan interpretation performed by a fellowship-trained, board-certified musculoskeletal radiologist following the BAMIC protocol.[25] HS or QD tears where the RTP date was unknown were excluded from this study.

Data Collection

This study was approved by the host institutional review board. The included athlete's sex, age, history of previous HS or QD injury (if recorded), date of injury, date of reinjury (if applicable), RTP date, laterality of injury, activity when injury occurred, and return to previous level of play were extracted from the athlete's electronic medical record (EMR). The EMR of included athletes was reassessed monthly to identify r-injury. MRI scan data obtained included: date of imaging, primary muscle affected, additional HS or QD muscles affected (if applicable), and BAMIC. MRI was performed on either a 1.5- or 3-T MRI scanner at a single imaging center at the host institution. All MRI scans included image sequences used to designate BAMIC.[25]

Protocol

The TRTP was calculated as the number of days between the date of index injury and the date that the athlete returned to football (practice, weightlifting, or competition) without any activity limitation enforced by the team medical staff. Day zero was defined as the date of index injury. The RTP date was defined as the date that a team athletic trainer or team physician changed the athlete's performance status to training and competition without any activity limitation.

Treatment and rehabilitation were provided by the team athletic trainers, physical therapists, and sports medicine physicians. Treatment consisted of rest, ice, elevation, pain management, and rehabilitation. BAMIC results were used for information purposes only and were not utilized to guide treatment. Rehabilitation was guided by the philosophy of increasing functional and strength-based recovery incorporating an adaptable return to running progression depending on the athlete's functional status and pain level. There were no specific criteria used for reintegration into practice, which occurred gradually, starting with individual, sport-specific drills, and slowly increasing exercise volume.

The decision for RTP was made by the team physician and athletic trainers. The decision for return to play was multifaceted and encompassed the athlete's subjective inclination to RTP, rehabilitation progress and clinical markers including defined parameters for bilateral lower extremity range of motion (ie, symmetric bilateral hip and knee range of motion), strength (ie, within 10% of symmetric knee flexion strength using a force

Table 1 Summary of British Athletics Muscle Injury Classification (* if any characteristics of a higher grade injury are present the injury is graded at the highest grade)

Grade	Description	MRI (day 2)
0a	Focal area of muscle pain usually following exercise	MRI normal
0b	Generalised muscle pain following unaccustomed exercise	MRI normal or patchy high signal change throughout one or more muscles
Grade	**Description**	**MRI**
1a	Small myofascial tear	High signal change evident at the fascial border with less than 10% extension into muscle belly Craniocaudal distance of <5 cm
1b	Small muscle-tendon junction tear	High signal change of less than 10% cross-sectional area of muscle at the muscle-tendon junction High signal change of craniocaudal length <5 cm (may note fibre disruption of <1 cm)
Grade	**Description**	**MRI**
2a	Moderate myofascial tear	High signal change evident at fascial border with extension into the muscle High signal change cross-sectional area of between 10% and 50% at maximal site High signal change of craniocaudal length >5 and <15 cm Architectural fibre disruption usually noted over less than 5 cm
2b	Moderate muscle-tendon junction tear	High signal change evident at the muscle-tendon junction High signal change cross-sectional area of between 10% and 50% at maximal site High signal change of craniocaudal length >5 and <15 cm Architectural fibre disruption usually noted over less than 5 cm
2c	Moderate sized intratendinous tear	High signal change extends into the tendon with longitudinal length of tendon involvement <5 cm Cross-sectional area of tendon involvement <50% of maximal tendon cross-sectional area No loss of tension or discontinuity within the tendon
Grade	**Description**	**MRI**
3a	Extensive myofascial tear	High signal change evident at fascial border with extension into the muscle High signal change cross-sectional area of greater than 50% at maximal site High signal change of craniocaudal length of greater than 15 cm Architectural fibre disruption usually noted over more than 5 cm
3b	Extensive muscle-tendon junction tear	High signal change cross-sectional area of greater than 50% at maximal site High signal change of craniocaudal length of greater than 15 cm Architectural fibre disruption usually noted over more than 5 cm
3c	Extensive intratendinous tear	High signal change extends into the tendon Longitudinal length of tendon involvement >5 cm Cross-sectional area of tendon involvement >50% of maximal tendon cross-sectional area There may be loss of tendon tension although no discontinuity is evident
Grade	**Description**	**MRI**
4	Full thickness tear of muscle	Complete discontinuity of the muscle with retraction
4c	Full thickness tear of tendon	Complete discontinuity of the tendon with retraction

Figure 1. The BAMIC (reproduced from Pollock et al[25]). BAMIC, British Athletics Muscle Injury Classification.

measurement sensor), resolution of pain, and incorporation of continuously assessed variables defined by global position systems (GPS). GPS measuring player load during practice (ie, combined metrics including sprint velocity, accelerations and decelerations, distance covered) were used as part of the clinical markers to determine RTP. Specifically, player load was gradually increased in practice, and, when an athlete's player load reached the average level for their position group, the athlete was considered healed according to GPS criteria.

A reinjury was recorded if the player sustained an acute, sudden exacerbation of pain in the same muscle and region as the index injury during exercise, followed by worsening of clinical tests including localized tenderness to palpation and changes with manual muscle testing and requiring cessation of current activity of rehabilitation or training for >48 hours.[25] All suspected reinjuries were evaluated by a team physician, who confirmed the diagnosis of reinjury. Reinjuries were included if the injury occurred during the index injury rehabilitation process or within 3 months of the day of an injured athlete was designated "as tolerated."

Statistical Analysis

Frequencies, mean, median, standard deviation, and interquartile range were used to describe participant/injury characteristics, TRTP, and RIR. Participant and injury characteristics were described for all 34 injuries among the 21 participants. However, as in Pollock et al,[25] an injury was excluded if the participant had a reinjury before returning to play for TRTP analysis. To assess differences in TRPT by BAMIC (0a-3b), grade (0-3), injury site (a-c), and muscle group, Kruskal-Wallis *H* and Mann-Whitney *U* tests were used where appropriate. For these analyses, repeat participants were excluded by selecting primary injury based on having a higher BAMIC. If 2 cases had the same BAMIC, then the injury to the long head of the bicep femoris was included. Median and interquartile range of TRTP were plotted by BAMIC, grade, injury site, and reinjury status. Separate linear regressions assessing TRTP by grade and injury site were conducted controlling for age and then for muscle group. All analyses were conducted using IBM SPSS Statistics (Version 29.0).

Table 1. Characteristics of study participants and HS and QD injuries

Participant (n = 21) and injured muscle characteristics	
Age, y[a]	20 (1.2)
Total injuries, n	34
Laterality	
Left, n (%)	15 (44.1)
Right, n (%)	19 (55.9)
Muscle group	
HS, n (%)	24 (70.6)
QD, n (%)	10 (29.4)
Muscle injured[b]	
Long head biceps femoris, n (%)	12 (35.3)
Short head biceps femoris, n (%)	3 (8.8)
Semimembranosus, n (%)	5 (14.7)
Semitendinosus, n (%)	4 (11.8)
Rectus femoris, n (%)	4 (11.8)
Vastus lateralis, n (%)	2 (5.9)
Vastus medialis, n (%)	3 (8.8)
BAMIC	
0a, n (%)	2 (5.9)
1a, n (%)	3 (8.8)
1b, n (%)	7 (20.6)
2a, n (%)	5 (14.7)
2b, n (%)	7 (20.6)
2c, n (%)	4 (11.8)
3a, n (%)	1 (2.9)
3b, n (%)	5 (14.7)
Reinjury among 21 athletes	
No, n (%)	17 81.0
Yes, n (%)	4 19.0
Reinjury before RTP, n (%)	3 75.0
Reinjury after RTP, n (%)	1 25.0
TRTP, days[c]	27 (18.8-34.5)

BAMIC, British Athletics Muscle Injury Classification; HS, hamstring; IQR, interquartile range; QD, quadriceps; RTP, return to play; TRTP, time to RTP.
[a]Age at the time of index injury if multiple injuries occurred; mean (SD).
[b]Percentages do not sum to 100% as >1 muscle can be injured.
[c]Median (IQR).

Table 2. TRTP and RIR of HS and QD injuries according to BAMIC (n = 30)

BAMIC	No. of Injuries	Median (IQR)[a]	Reinjury, n (%)
0a	2	9 (NA[b])	0 (0)
1a	3	24 (NA[b])	1 (33.0)
1b	7	33 (27.0, 34.0)	0 (0)
2a	4	29 (19.5, 35.3)	0 (0)
2b	6	26 (15.8, 45.0)	1 (33.0)
2c	4	31 (20.5, 33.3)	0 (0)
3a	1	23 (NA[b])	0 (0)
3b	3	26 (NA[b])	0 (0)

BAMIC, British Athletics Muscle Injury Classification; HS, hamstring; ; IQR, interquartile range NA, not applicable; QD, quadriceps; RTP, return to play; TRTP, time to RTP.
[a]Four reinjuries that occurred before full RTP were excluded from the median TRTP calculations (2a [1], 2b [1], 3b [2]).
[b]Not possible to calculate IQR (n = ≤3).

RESULTS

Study Participants and Injury Characteristics

The study participant and injured muscle characteristics are described in Table 1. Athletes were excluded if there was no identifiable RTP date (n = 3), if an MRI scan was not obtained (n = 2), and due to incorrect diagnosis (ie, femoral diaphyseal stress fracture, n = 1). A total of 11 (52.0%) of the athletes suffered ≥2 separate muscle injuries during the index injury. One athlete suffered an index injury involving both his HS and QD. Most (89.3%) tears occurred before the season and all tears occurred in practice. All injured athletes returned to previous level of play.

Association Between TRTP, BAMIC, and Muscle Group

The median TRTP and number of injuries by BAMIC are presented in Table 2. Most injuries were classified as BAMIC 2a-3b (60.0%) and 2 injuries (<0.1%) were 0a. The median TRTP for muscle injuries according to BAMIC, grade, and injury site are presented in Figure 2. The median TRTP was 26 days (interquartile range [IQR], 17.8-33.0) with BAMIC 0a included and 29 days (IQR, 23.3-35.5) with BAMIC 0a excluded, which translates to approximately 4 weeks of missed play due to injury.

Kruskal-Wallis *H* tests of unique patients showed no statistically significant relationship between TRTP by BAMIC (H = 0.9; $P = 0.99$; n = 18; $\varepsilon^2 = 0.055$), grade (H = 0.6; $P = 0.76$; n = 18; $\varepsilon^2 = 0.032$), or injury site (H = 0.1; $P = 0.94$; n = 18 $\varepsilon^2 = 0.008$) (Table 3). However, TRTP for grade 0 injuries was less compared with grade 1 to 3 injuries, and TRTP was greater for injury site "c" or intratendinous injuries compared with "a" and "b" site injuries (Figure 2).

There was no association between median TRTP for HS (n = 15) versus QD injuries (n = 5) ($Z = -0.7$; $P = 0.48$; $\varepsilon^2 = 0.042$) (Table 3). Separate linear regression analyses revealed no significant association between injury site (B = 1.3, confidence interval [CI], –7.1 to 9.7; $P = 0.75$; $\beta = 0.083$) or grade (B = 0.3; CI, –8.6 to 9.2; $P = 0.95$; $\beta = 0.017$) and TRTP when controlling for age. Likewise, there was no significant association between injury site (B = 1.4; CI, –6.7 to 9.6; $P = 0.72$; $\beta = 0.091$) or grade (B = 0.4; CI, –8.2 to 9.1; $P = 0.91$; $\beta = 0.027$) and TRTP when controlling for muscle group.

Association Between RIR, BAMIC, and Muscle Group

The summary of reinjuries is reported in Figure 3. Most (75.0%) reinjuries occurred before RTP, were grade 2, and were "b" site injuries. The median number of days between the index injury and the reinjury for athletes who suffered a reinjury before returning to play was 25 days (range, 20-46 days) and for athletes who suffered a reinjury after returning to play was 120 days. The median TRTP for reinjury status are presented in Figure 4. We were unable to assess the statistical significance between BAMIC, grade, or injury site and RIR due to the low number of reinjuries.

DISCUSSION

Summary of Principal Findings

This study substantiated that activity-related HS and QD tears are common in American football college athletes, particularly in practice and preseason, and lead to considerable time lost to participation. There was no statistically significant relationship or meaningful effect sizes between BAMIC or muscle group and

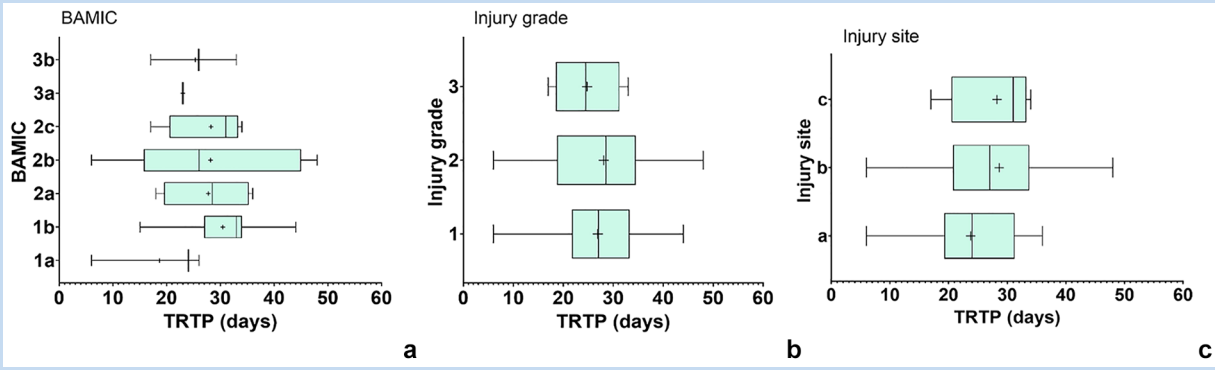
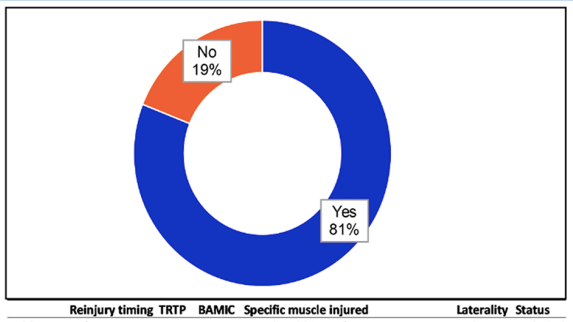


Figure 2. (a) BAMIC and subclassifications of muscle injuries by (b) injury grade and (c) injury site. Data are shown as median (IQR, range) TRTP (days).
BAMIC, British Athletics Muscle Injury Classification; IQR, interquartile range; RTP, return to play; TRTP, time to RTP.

Table 3. Comparison of BAMIC and TRTP (n = 18)

	No. of Athletes	Median (IQR)[a]	Kruskal-Wallis *H*	*P* Value	Effect Size
BAMIC			0.9	0.99	0.055
1a	1	26 (NA[b])			
1b	2	21 (NA[b])			
2a	3	24 (NA[b])			
2b	5	26 (13.0, 46.0)			
2c	3	31 (NA[b])			
3a	1	23 (NA[b])			
3b	3	26 (NA[b])			
Grade			0.6	0.76	0.032
1	3	26 (NA)			
2	11	26 (18, 36)			
3	4	24.5 (19, 31)			
Site			0.1	0.94	0.008
a	5	24 (21, 31)			
b	10	26 (17, 36)			
c	3	31 (NA)			
Muscle group			$Z = -0.8$	0.52	0.042
HS	13	25 (18, 35)			
QS	5	26 (11, 30)			

Sample is 19 for analyses where grade 0 was excluded. Removes repeat patient by selecting primary highest-grade injury per BAMIC. If an athlete had 2 injuries with the same BAMIC grade, then the long head biceps femoris was selected. BAMIC, British Athletics Muscle Injury Classification; HS, hamstring; IQR, interquartile range; NA, not applicable; QD, quadriceps; RTP, return to play; TRTP, time to RTP.
[a]Four reinjuries that occurred before full RTP were excluded from the median TRTP calculations (2a [1], 2b [1], 3b [2]).
[b]Not possible to calculate IQR (n = ≤3).



	Reinjury timing	TRTP	BAMIC	Specific muscle injured	Laterality	Status
Athlete 1	Before	58	2a	Long head biceps femoris	Right	Excluded
Athlete 2	After	7	2b	Rectus femoris & quadriceps	Right	Included
Athlete 3	Before	77	3b	Long head biceps femoris & hamstring	Right	Excluded
Athlete 4	Before	53	2b	Long head biceps femoris	Right	Excluded

Figure 3. Summary of reinjuries among athletes.
BAMIC, British Athletics Muscle Injury Classification; RRTP, return to play; TRTP, time to RTP.

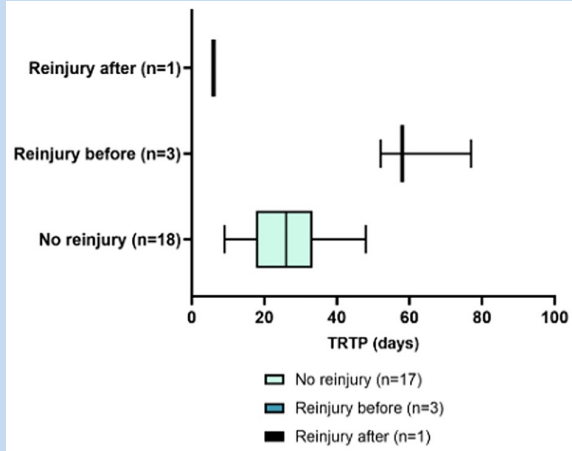


Figure 4. Median (IQR, range) TRTP (days) to return to play by reinjury status.
BAMIC, British Athletics Muscle Injury Classification; IQR, interquartile range; RTP, return to play; TRTP, time to RTP.

TRTP. Similarly, BAMIC grade and injury site lacked significance and meaningful effect sizes. However, upon visual inspection, there was greater TRTP for injury site "c" or intratendinous injuries compared with "a" and "b" injuries and TRTP was less for grade 0 injuries compared with grade 1 to 3 injuries. Most athletes missed about 4 weeks of play, irrespective of BAMIC, albeit with nearly 1 in 5 athletes suffering a reinjury. The low number of reinjuries precluded assessment for a relationship between BAMIC and RIR. Next, we will discuss the findings related to BAMIC, including TRTP, RIR, and muscle characteristics identified in this study in further detail.

BAMIC and TRTP

A significant relationship has been identified between BAMIC grade and TRTP, indicating that injuries of increasing grade have longer TRTP.[20,24,25,29] Specifically, BAMIC grade 0 injuries, also considered MRI "negative" injuries, are associated with shorter TRTP.[6,17,24] The athlete in this study who sustained a grade 0 injury had a TRTP of 9 days, aligning with previous data and clinically resulting in significantly fewer missed games than the TRTP for the remaining athletes in this study. Yet, there was no statistically significant relationship between BAMIC and TRTP in this study.

Compared with muscle belly/myotendinous junction injuries (BAMIC injury site "b"), myofascial injuries (BAMIC injury site "a") have demonstrated the shortest TRTP, whereas intratendinous injuries (BAMIC injury site "c") have shown the longest TRTP.[17,24-26] This study revealed greater TRTP for injury site "c" injuries, with a mean TRTP of 31 days (range, 21-33 days), compared with shorter TRTP for injury site "a" injuries, with a mean TRTP of 24 days (range, 19-31 days). Clinically, this difference in TRTP between injury site "a" and "c" injuries in this study represents the variance of missing an additional game. However, the practical significance of these findings is limited due to the small effect sizes in this study.

BAMIC and RIR

Due to the low number of reinjuries in this study, the sample size precluded assessment of the relationship between BAMIC and RIR. For the athletes who suffered a reinjury after returning to play, the injury involved the QD and were classified as 1a and 2b. For the athletes who experienced a reinjury before returning to play, all involved the HS and were classified as 2a (n = 1), 2b (n = 1), and 3b (n = 2). However, previous studies suggest a relationship between BAMIC and RIR, ie, that increasing BAMIC grade and injury site "c" injuries have been associated with a greater risk of reinjury.[18,25,28]

Muscle Injury Characteristics

There was no difference in TRTP for HS and QD injuries in this study ($P = 0.48$). Comparably, a study of HS injuries and QD injuries in NFL athletes found similar recovery times,[12] although the median TRTP for BAMIC grade 1 to 3 injuries in this study was approximately twice that reported by Herzog et al,[12] possibly due to the low number of BAMIC grade 0 injuries in this study. Also, when controlling for age or muscle group (ie, HS or QD), there was no association between BAMIC and TRTP. Although these findings may have been expected due to the lack of overall significance for BAMIC and TRTP, the potential difference in TRTP between HS and QD injuries as predicted by BAMIC remains relevant given the prevalence of HS and QD injures in football.

Limitations

Our study cohort represented a small number of HS and QD injuries in 1 NCAA-DI college football team. A limitation to this study is the sample size and inherent underpowering, particularly the lack of a sufficient number of injuries in each BAMIC category, which limits the definitive ability to establish conclusions. Primarily, there was 1 grade 0 injury and zero grade

4 injuries. In addition, there were an insufficient number of reinjuries to assess for significance between BAMIC and RIR. Due to the significant burden of lower extremity muscle injuries and the limited ability of current muscle injury classification and grading systems to accurately predict clinical outcomes including TRTP and RIR, further studies with larger sample sizes are necessitated. Previous authors have called for a multiteam, multileague approach to increase the sample size of studies evaluating lower extremity muscle injuries in athletes.[18,28] Given the frequency of lower extremity muscle injuries in football and college athletes, a similar approach in these populations could be considered. Because the team medical staff was not blinded to the BAMIC of an athlete's injury when making rehabilitation and RTP decisions, this presents a potential source of bias. Finally, patient care is not based solely on imaging. Although this study describes the prognostic value of an imaging classification, it is important to recognize that a patient's physical examination is equally crucial in the overall clinical decision-making process.

CONCLUSION

This is the first appraisal of BAMIC in college and American football athletes. The notable prevalence of muscle tears in our cohort highlights the need for prevention, particularly in the preseason and early in the regular season. Although BAMIC lacked clinical value in this study, the lack of enough injuries in each category limits our findings. Future studies are needed to assess the value of BAMIC in college and football athletes.

DATA ACCESSIBILITY STATEMENT

All data are available as requested through the study principal investigator.

ORCID ID

William Hollabaugh https://orcid.org/0000-0001-7064-327X

REFERENCES

1. Alonso JM, Edouard P, Fischetto G, Adams B, Depiesse F, Mountjoy M. Determination of future prevention strategies in elite track and field: analysis of Daegu 2011 IAAF Championships injuries and illnesses surveillance. *Br J Sports Med*. 2012;46(7):505-514.
2. Biglands JD, Grainger AJ, Robinson P, et al. MRI in acute muscle tears in athletes: can quantitative T2 and DTI predict return to play better than visual assessment? *Eur Radiol*. 2020;30(12):6603-6613.
3. Brooks JH, Fuller CW, Kemp SP, Reddin DB. Incidence, risk, and prevention of hamstring muscle injuries in professional rugby union. *Am J Sports Med*. 2006;34(8):1297-1306.
4. Chandran A, Morris SN, Powell JR, Boltz AJ, Robison HJ, Collins CL. Epidemiology of injuries in National Collegiate Athletic Association men's football: 2014-2015 through 2018-2019. *J Athl Train*. 2021;56(7):643-650.
5. Cohen SB, Towers JD, Zoga A, et al. Hamstring injuries in professional football players: magnetic resonance imaging correlation with return to play. *Sports Health*. 2011;3(5):423-430.
6. Ekstrand J, Askling C, Magnusson H, Mithoefer K. Return to play after thigh muscle injury in elite football players: implementation and validation of the Munich muscle injury classification. *Br J Sports Med*. 2013;47(12):769-774.
7. Ekstrand J, Hagglund M, Walden M. Epidemiology of muscle injuries in professional football (soccer). *Am J Sports Med*. 2011;39(6):1226-1232.
8. Feddermann-Demont N, Junge A, Edouard P, Branco P, Alonso JM. Injuries in 13 international Athletics championships between 2007-2012. *Br J Sports Med*. 2014;48(7):513-522.
9. Feeley BT, Kennelly S, Barnes RP, et al. Epidemiology of National Football League training camp injuries from 1998 to 2007. *Am J Sports Med*. 2008;36(8):1597-1603.
10. Freckleton G, Pizzari T. Risk factors for hamstring muscle strain injury in sport: a systematic review and meta-analysis. *Br J Sports Med*. 2013;47(6):351-358.
11. Hamilton B, Valle X, Rodas G, et al. Classification and grading of muscle injuries: a narrative review. *Br J Sports Med*. 2015;49(5):306.
12. Herzog MM, Weiss L, Lee RY, et al. Lower extremity strains in the US National Football League, 2015-2019. *Am J Sports Med*. 2023;51(8):2176-2185.
13. Hollabaugh WL, Sin A, Walden RL, et al. Outcomes of activity-related lower extremity muscle tears after application of the British Athletics Muscle Injury Classification: a systematic review. *Sports Health*. 2024;16(5):783-796.
14. Kumaravel M, Bawa P, Murai N. Magnetic resonance imaging of muscle injury in elite American football players: predictors for return to play and performance. *Eur J Radiol*. 2018;108:155-164.
15. Luxenburg D, Wasserman NA, Geller JS, Perez JR, Burke J, Kaplan LD. Hamstring injuries in the national football league: an epidemiological study. *J Orthop*. 2023;35:111-114.
16. Macdonald B, McAleer S, Kelly S, Chakraverty R, Johnston M, Pollock N. Hamstring rehabilitation in elite track and field athletes: applying the British Athletics Muscle Injury Classification in clinical practice. *Br J Sports Med*. 2019;53(23):1464-1473.
17. McAleer S, Macdonald B, Lee J, et al. Time to return to full training and recurrence of rectus femoris injuries in elite track and field athletes 2010-2019; a 9-year study using the British Athletics Muscle Injury Classification. *Scand J Med Sci Sports*. 2022;32(7):1109-1118.
18. McAuley S, Dobbin N, Morgan C, Goodwin PC. Predictors of time to return to play and re-injury following hamstring injury with and without intramuscular tendon involvement in adult professional footballers: a retrospective cohort study. *J Sci Med Sport*. 2022;25(3):216-221.
19. Mueller-Wohlfahrt HW, Haensel L, Mithoefer K, et al. Terminology and classification of muscle injuries in sport: the Munich consensus statement. *Br J Sports Med*. 2013;47(6):342-350.
20. Nescolarde L, Terricabras J, Mecho S, Rodas G, Yanguas J. Differentiation between tendinous, myotendinous and myofascial injuries by L-BIA in professional football players. *Front Physiol*. 2020;11:574124.
21. Pezzotta G, Pecorelli A, Querques G, Biancardi S, Morzenti C, Sironi S. MRI characteristics of adductor longus lesions in professional football players and prognostic factors for return to play. *Eur J Radiol*. 2018;108:52-58.
22. Pezzotta G, Querques G, Pecorelli A, Nani R, Sironi S. MRI detection of soleus muscle injuries in professional football players. *Skeletal Radiol*. 2017;46(11):1513-1520.
23. Pollock N, James SL, Lee JC, Chakraverty R. British athletics muscle injury classification: a new grading system. *Br J Sports Med*. 2014;48(18):1347-1351.
24. Pollock N, Kelly S, Lee J, et al. A 4-year study of hamstring injury outcomes in elite track and field using the British Athletics rehabilitation approach. *Br J Sports Med*. 2022;56(5):257-263.
25. Pollock N, Patel A, Chakraverty J, Suokas A, James SL, Chakraverty R. Time to return to full training is delayed and recurrence rate is higher in intratendinous ('c') acute hamstring injury in elite track and field athletes: clinical application of the British Athletics Muscle Injury Classification. *Br J Sports Med*. 2016;50(5):305-310.
26. Shamji R, James SLJ, Botchu R, Khurniawan KA, Bhogal G, Rushton A. Association of the British Athletic Muscle Injury Classification and anatomic location with return to full training and reinjury following hamstring injury in elite football. *BMJ Open Sport Exerc Med*. 2021;7(2):e001010.
27. Soligard T, Steffen K, Palmer D, et al. Sports injury and illness incidence in the Rio de Janeiro 2016 Olympic Summer Games: a prospective study of 11274 athletes from 207 countries. *Br J Sports Med*. 2017;51(17):1265-1271.
28. Tears C, Rae G, Hide G, et al. The British Athletics Muscle Injury Classification grading system as a predictor of return to play following hamstrings injury in professional football players. *Phys Ther Sport*. 2022;58:46-51.
29. Wangensteen A, Guermazi A, Tol JL, et al. New MRI muscle classification systems and associations with return to sport after acute hamstring injuries: a prospective study. *Eur Radiol*. 2018;28(8):3532-3541.

For article reuse guidelines, please visit Sage's website at http://www.sagepub.com/journals-permissions.